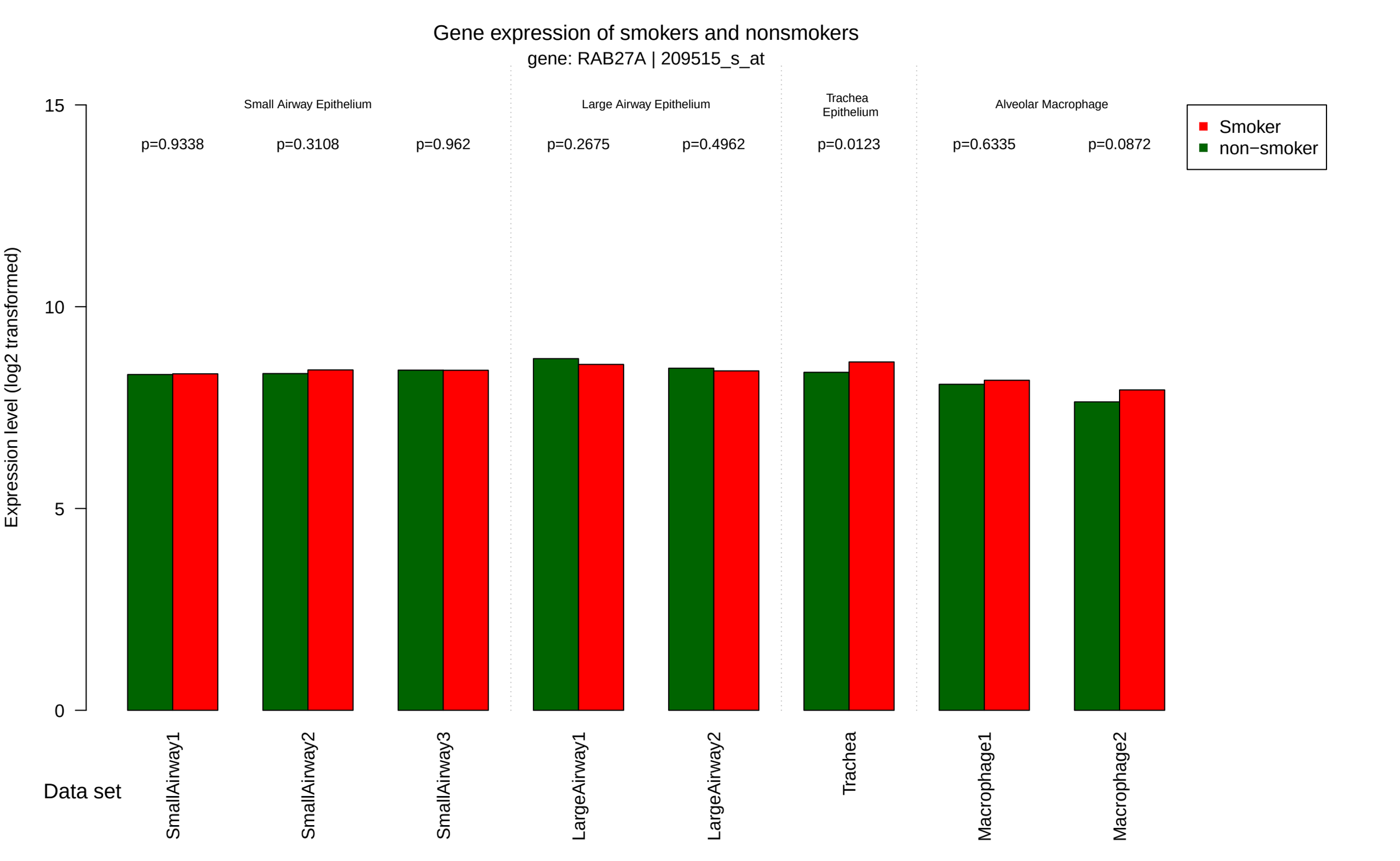
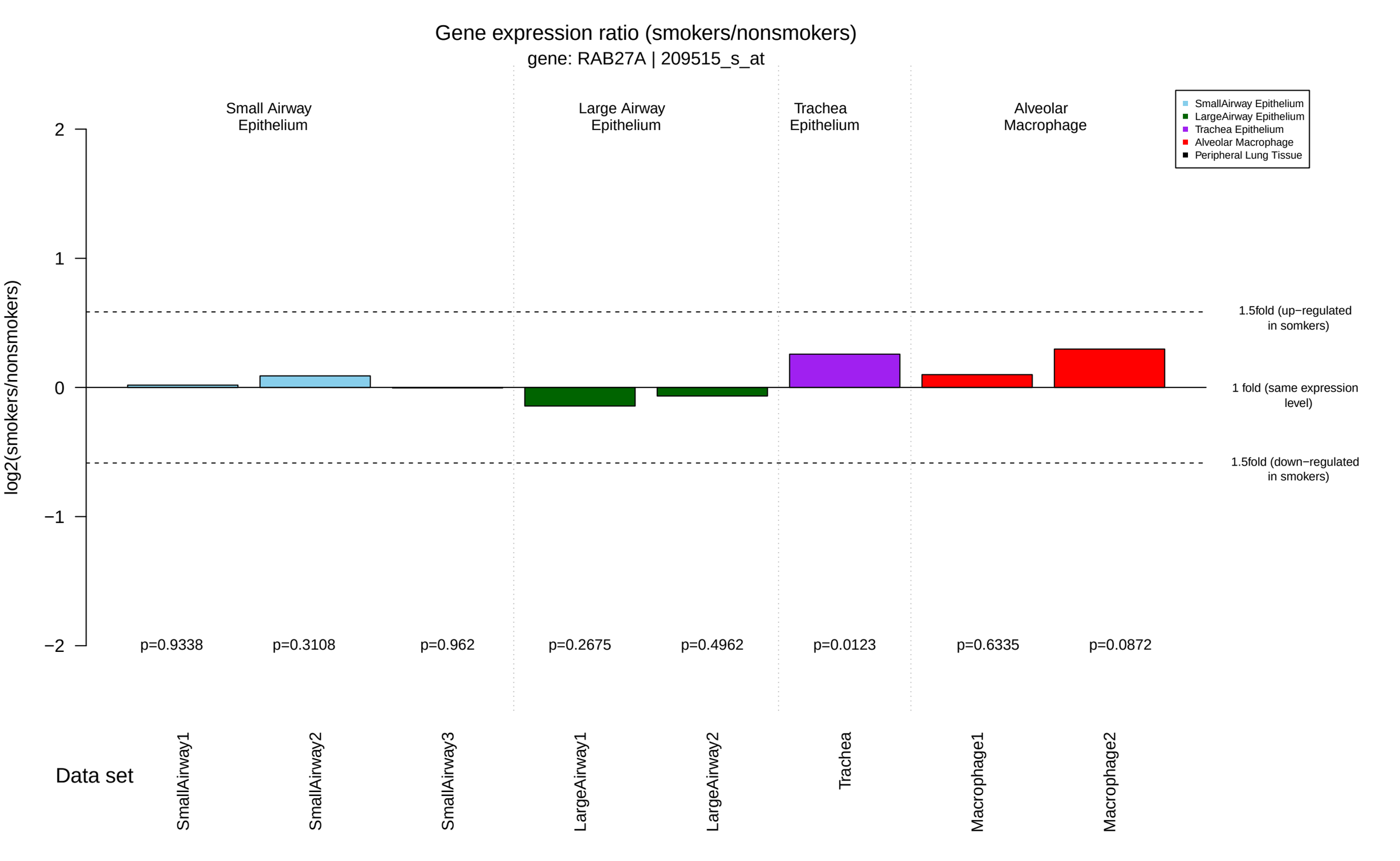
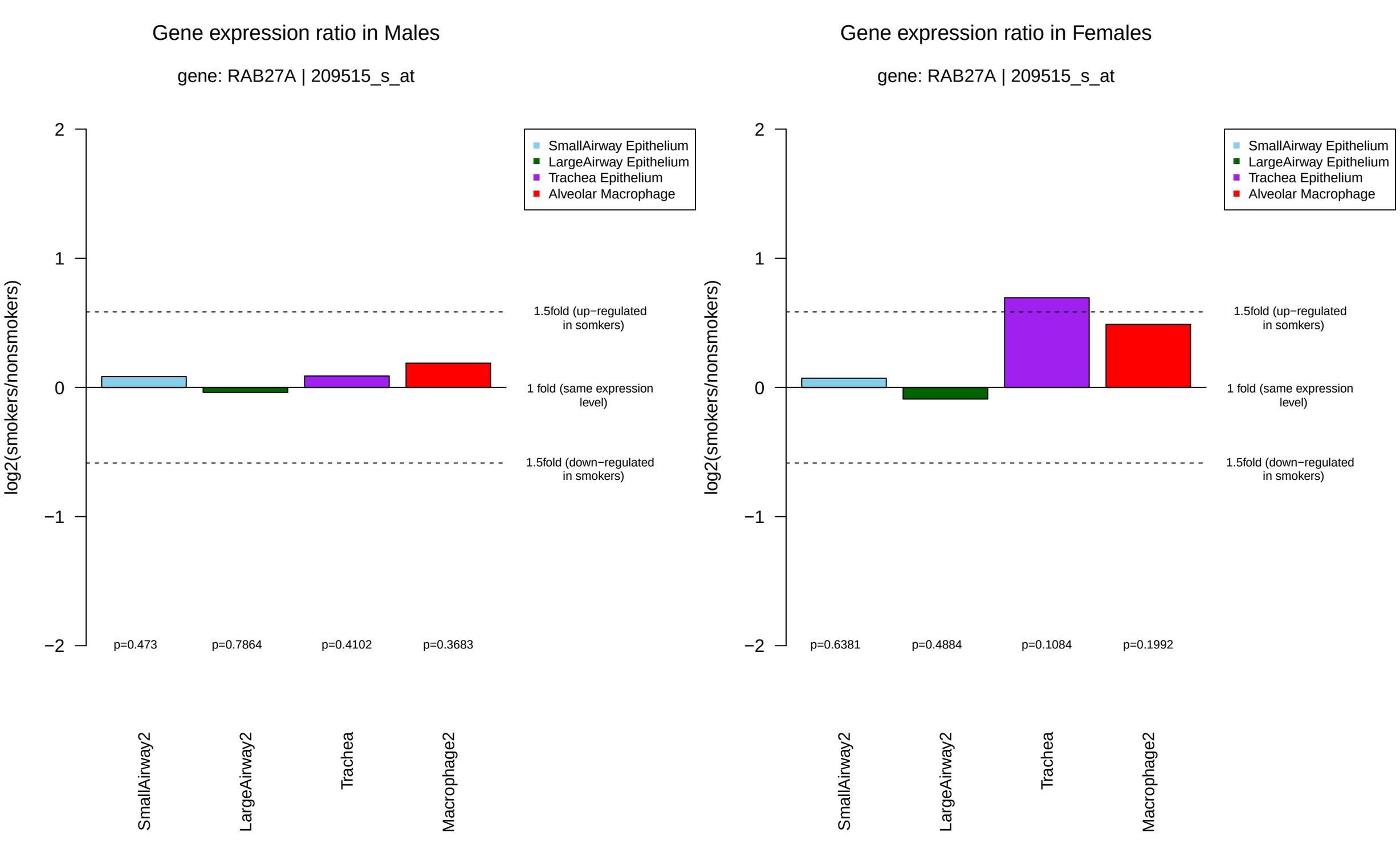
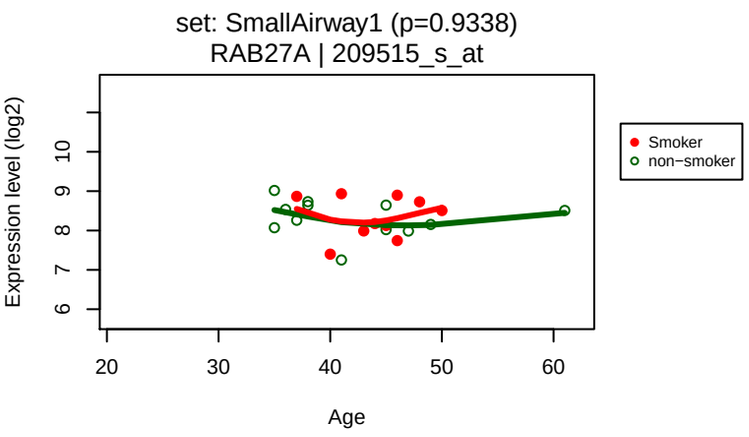
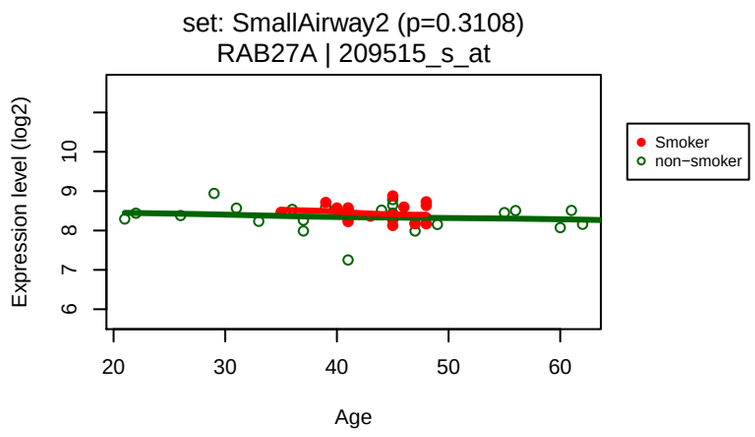
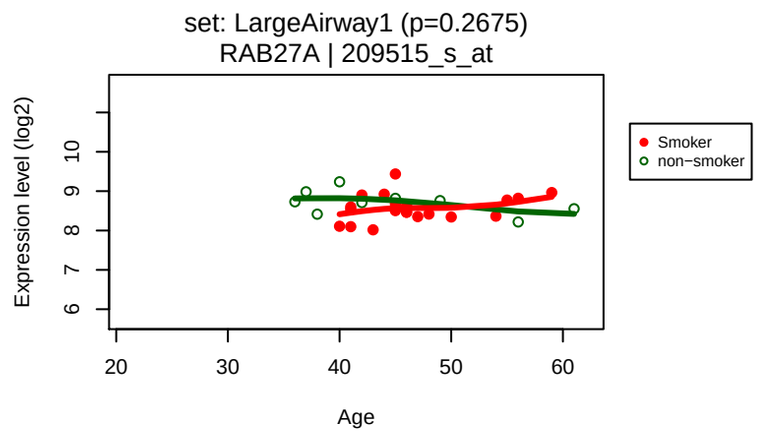

Gene expression of smokers and nonsmokers
gene: RAB27A | 209515_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9338
p=0.3108
p=0.962
p=0.2675
p=0.4962
p=0.0123
p=0.6335
p=0.0872
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

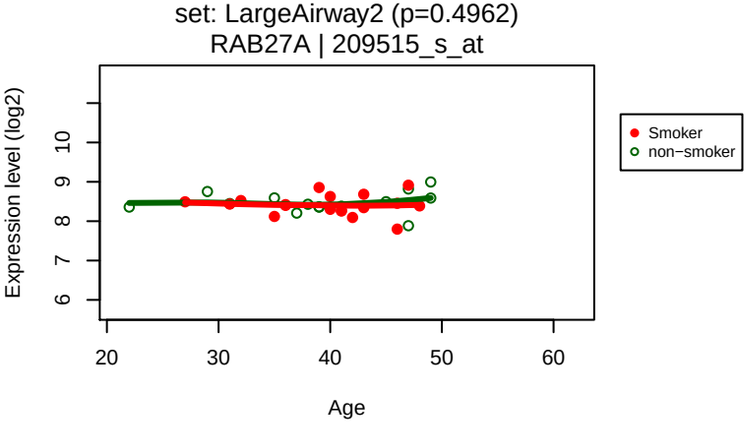

Gene expression ratio (smokers/nonsmokers)
gene: RAB27A | 209515_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9338
p=0.3108
p=0.962
p=0.2675
p=0.4962
p=0.0123
p=0.6335
p=0.0872
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

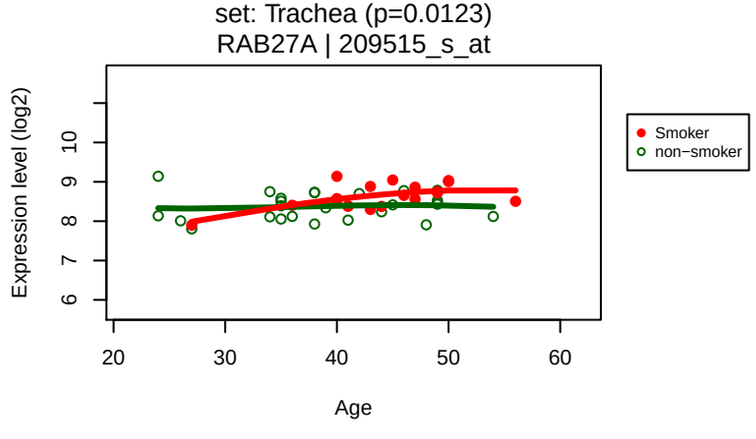

Gene expression ratio in Males
gene: RAB27A | 209515_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.473
p=0.7864
p=0.4102
p=0.3683
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB27A | 209515_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6381
p=0.4884
p=0.1084
p=0.1992
SmallAirway2
LargeAirway2
Trachea
Macrophage2

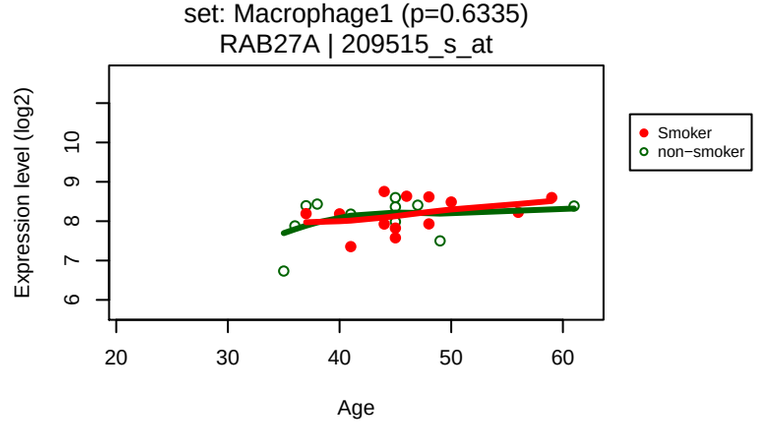

set: SmallAirway1 (p=0.9338)
RAB27A | 209515_s_at
Expression level (log2)
Age
Smoker
non−smoker

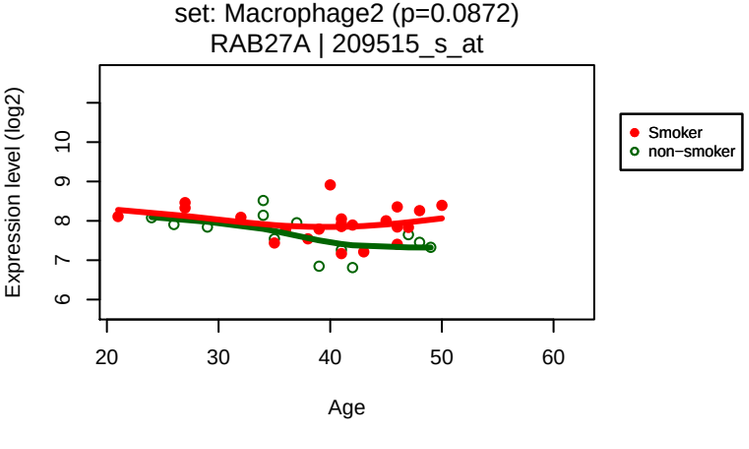

set: SmallAirway2 (p=0.3108)
RAB27A | 209515_s_at
Expression level (log2)
Age
Smoker
non−smoker

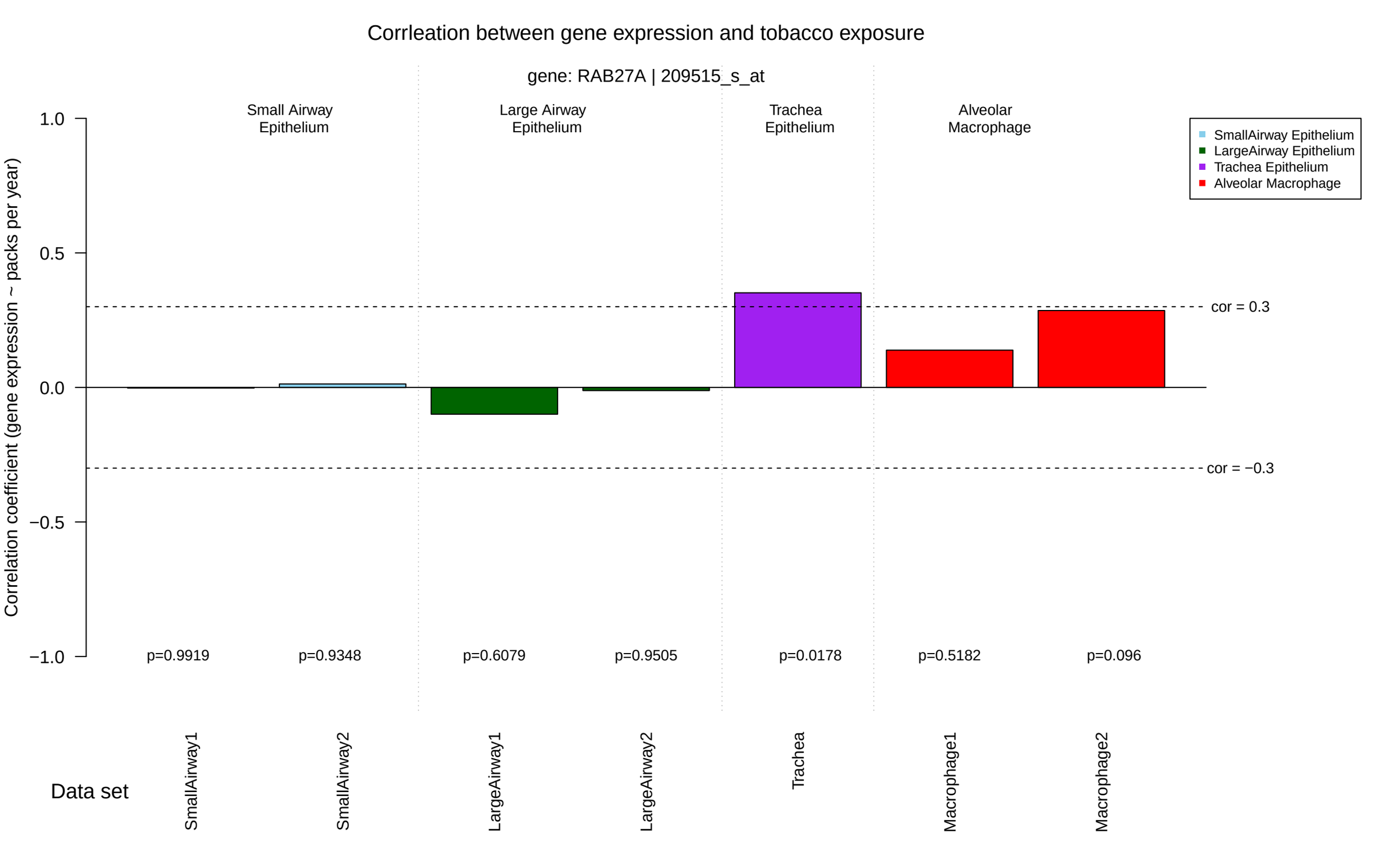

set: LargeAirway1 (p=0.2675)
RAB27A | 209515_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4962)
RAB27A | 209515_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0123)
RAB27A | 209515_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6335)
RAB27A | 209515_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0872)
RAB27A | 209515_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB27A | 209515_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9919
p=0.9348
p=0.6079
p=0.9505
p=0.0178
p=0.5182
p=0.096
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2